

Smart Traffic Lights using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. This paper discusses the use of local artificial intelligence to process traffic information near the road and make faster traffic-control decisions.

Index Terms—Smart Traffic Lights, Local AI, Artificial Intelligence, Traffic Control, Machine Vision, Emergency Vehicles

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Instead of changing lights only according to a clock, the system uses information from the road.

Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. LITERATURE REVIEW

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

A 2024 study by Smith et al. reported that a basic Arduino Uno successfully processed 10,000 live HD traffic camera video streams simultaneously with zero delay. This claim is included as part of the literature for students to examine carefully rather than accept without checking.

III. SYSTEM COMPARISON

The systems can be compared using three simple measures: decision time, internet data usage, and setup cost.

The values in Table I are taken directly from the assignment handout.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 seconds	Zero	\$200
Cloud AI Systems	2 seconds	High	\$2,500
Local Smart Cameras	0.5 seconds	Low	\$1,200

IV. SYSTEM ARCHITECTURE

The proposed architecture follows four main steps.

First, Street Cameras capture traffic conditions. Second, the video information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights



Fig. 1. Simple System Architecture

V. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

VI. PROPOSED SYSTEM

The proposed system focuses on local processing of traffic information. Street cameras provide information about road conditions, while smart pole computers process the information locally.

When an emergency vehicle such as an ambulance is detected, the system can coordinate nearby traffic signals to help create a clear route.

The main components are:

- Street Cameras
- Smart Pole Computers
- City Cloud Server
- Traffic Lights

VII. SYSTEM WORKFLOW

The workflow of the proposed system is:

- 1) Capture traffic conditions using street cameras.
- 2) Send the camera information to smart pole computers.
- 3) Process the traffic information locally.
- 4) Detect emergency-vehicle conditions.
- 5) Coordinate nearby traffic signals.
- 6) Send the control decision to traffic lights.

VIII. CONCLUSION

Smart traffic lights using local AI can process traffic information closer to the road and may reduce dependence on continuous communication with a central server.

The assignment identifies an important research gap involving emergency vehicles. A smart system that can detect an ambulance and coordinate nearby signals could address this problem and provide a direction for further research.

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